

**GREENCLOSET: SMART HUMIDITY & MOLD DETECTION FOR
SUSTAINABLE WARDROBES**

21CSE253T/ INTERNET OF THINGS

CASE STUDY REPORT

Submitted by

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Certified that the Case Study Report titled **GREENCLOSET: SMART HUMIDITY & MOLD DETECTION FOR SUSTAINABLE WARDROBES** is a bonafide work of **SHERENE FAUSTINA A (RA2311003020482)**, who carried out the case study work under the guidance of Ms.R.Lakshmi, Assistant Professor, CSE at SRM Institute of Science and Technology, Ramapuram. This case study work confirms to 21CSE253T/Internet Of Things, IV Semester, II year, 2024.

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I hereby declare that the entire work contained in this case study report titled **“GREENCLOSET:SMART HUMIDITY & MOLD DETECTION FOR SUSTAINABLE WARDROBES”** is carried out by **SHERENE FAUSTINA A (RA2311003020482)** at SRM Institute of Science and Technology, Ramapuram, Chennai, under the guidance of **Ms.R.LAKSHMI**, Assistant Professor, Department of Computer Science and Engineering.

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ABSTRACT

In an era where sustainable living has become not just a trend but a necessity, GreenCloset emerges as an innovative intersection between IoT technology and conscious fashion care. This case study introduces an intelligent system that monitors humidity levels and potential mold growth inside wardrobes — a space often overlooked in terms of environmental control yet highly susceptible to damaging microclimates. By leveraging real-time data collection and smart alerts, GreenCloset aims to enhance the longevity of clothing while reducing waste caused by fabric deterioration.

The system integrates humidity and temperature sensors with a microcontroller to continuously evaluate environmental conditions within the wardrobe. If unsafe moisture levels are detected — levels that typically promote fungal growth — the system promptly notifies the user through an intuitive mobile interface or buzzer alert. To support proactive wardrobe care, the system may also trigger auxiliary components such as mini dehumidifiers or ventilation fans, ensuring optimal storage conditions without manual intervention.

Beyond functionality, GreenCloset promotes sustainable fashion behavior by preserving fabric integrity and encouraging users to rethink their daily wardrobe care routines. The solution is designed to be energy-efficient, cost-effective, and scalable, making it accessible for eco-conscious households worldwide. It not only prevents material damage but also contributes to reducing the ecological footprint of fast fashion by increasing the lifespan of garments.

Through the fusion of IoT and environmental responsibility, GreenCloset redefines the way we interact with our personal spaces. It reflects a shift towards smarter, greener living — where even our closets participate in creating a healthier planet.

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LIST OF ACRONYMS AND ABBREVIATIONS

IoT	Internet of Things
AI	Artificial Intelligence
DHT	Digital Humidity & Temperature Sensor
ESP32	Espressif Systems Platform
LED	Light Emitting Diode
Wi-Fi	Wireless Fidelity
MCU	Microcontroller Unit
SMS	Short Message Service

CHAPTER 1

INTRODUCTION



Fig.1.1 Typical Household Wardrobe Susceptible to Humidity and Mold

In an era defined by technological advancement and environmental urgency, the intersection of sustainability and smart living has never been more vital. While innovations like smart lighting, security systems, and energy meters have reshaped modern homes, one area continues to operate in a state of analog neglect — our wardrobes. Though seemingly mundane, closets and clothing storage units serve a crucial purpose in everyday life, yet they remain passive spaces vulnerable to environmental damage, particularly from humidity and temperature fluctuations. The lack of attention to this micro-environment has led to the silent degradation of garments and the unwarranted rise in textile waste.

Enter GreenCloset — an intelligent, IoT-enabled solution aimed at transforming the conventional wardrobe into a smart, self-aware storage ecosystem. Designed with sustainability at its core, GreenCloset is more than a device; it is a holistic system that combines environmental monitoring, smart alerting, and behavioral nudging to protect garments, promote eco-conscious decisions, and extend the lifecycle of clothing.

The Rationale Behind GreenCloset

The fashion industry, while a beacon of self-expression, is one of the world's most resource-intensive sectors. From water-intensive cotton farming to synthetic fiber

pollution, the environmental footprint of fashion spans the globe. Yet, a less spoken contributor to this problem is post-purchase waste — clothes that are prematurely discarded not due to fashion obsolescence, but because of physical damage caused by poor storage, mildew, or humidity-induced degradation. This issue is especially prevalent in regions with tropical or coastal climates, where high ambient moisture is a year-round concern.

Traditional storage solutions offer little protection. Silica gel packets, naphthalene balls, and dehumidifiers provide temporary fixes but lack real-time intelligence and adaptability. GreenCloset seeks to address this neglected aspect of textile sustainability through a fusion of affordable technology and conscious design.

System Architecture – Dissecting the Components of GreenCloset

The GreenCloset system is thoughtfully engineered, modular in design, and highly adaptable to various storage environments. The following sections describe each component of the system in detail:

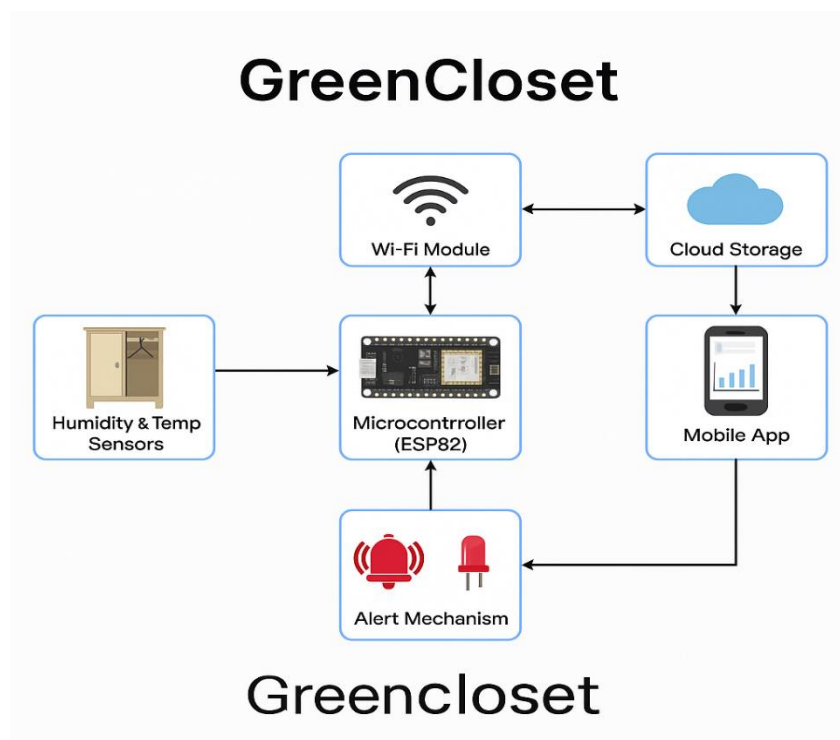


Fig.1.2 System Architecture

1. DHT22/AM2302 – Humidity & Temperature Sensor

At the heart of GreenCloset is its environmental sensing capability. The DHT22 sensor, also known as AM2302, is a high-accuracy, low-cost digital sensor capable of measuring both relative humidity and temperature. It provides real-time data at regular intervals and operates with minimal power consumption, making it ideal for embedded systems.

- **Humidity Range:** 0–100% RH (± 2 –5% accuracy)
- **Temperature Range:** -40 to +80°C (± 0.5 °C accuracy)
- **Why it matters:** These metrics are critical for determining when wardrobe conditions become conducive to mold formation. A spike in relative humidity beyond 60%, especially at room temperature, can trigger mildew growth.

2. Microcontroller Unit (MCU) – ESP8266 / ESP32

The ESP8266 or ESP32 acts as the brain of the GreenCloset system. These Wi-Fi-enabled microcontrollers are compact, cost-effective, and highly suitable for IoT applications.

- **Role:** The microcontroller collects data from the sensors, processes the readings, and determines when an environmental threshold has been breached.
- **Connectivity:** Using built-in Wi-Fi capabilities, the MCU communicates with a mobile app or cloud service, enabling remote monitoring.
- **Low Power:** Designed for continuous operation on minimal energy, supporting the project's eco-conscious mission.

3. Alert System – LED Indicators, Buzzers, and Notifications

To ensure immediate user response, GreenCloset integrates multiple layers of alerts:

- **LED Lights:** Visual indication of safe or unsafe conditions (e.g., Green for safe, Red for high humidity).

- **Buzzers:** Audible alarms triggered when humidity or temperature exceeds safe levels.
- **Mobile Notifications:** Through a mobile application or SMS gateway, users receive real-time alerts, ensuring action even if they're away from home.

4. **Cloud Storage and Dashboard**

GreenCloset is future-ready with options for cloud connectivity using platforms like Firebase, Blynk, or ThingSpeak.

- **Data Logging:** Historical data can be stored to observe patterns and seasonal variations.
- **Visualization:** Users can view live environmental graphs, monitor multiple wardrobes, and analyze trends.
- **Scalability:** This setup is extendable to multiple rooms, closets, or even community laundromats.

5. **Power Management – Battery or USB-powered**

- **USB Supply:** Ideal for permanent wardrobe installations.
- **Rechargeable Batteries:** For flexibility, portability, and resilience during power outages. Solar-powered versions may be explored for off-grid applications in future iterations.

6. **Optional Extensions & Future Integrations**

GreenCloset's open architecture supports multiple expansion modules:

- **Automatic Ventilation Control:** Integration with servo motors or exhaust fans to regulate airflow automatically.
- **Mobile App Control:** Custom-built applications for Android/iOS to control, monitor, and receive real-time updates.
- **AI-Driven Mold Prediction:** Using machine learning to anticipate mold risks based on historic environmental data and seasonal climate trends.
- **Voice Assistant Integration:** Compatibility with Google Assistant, Alexa, or Siri for voice-based queries (e.g., "Is my closet safe right now?").

- **AR Visualization:** Display wardrobe analytics via augmented reality overlays for next-gen smart home interfaces.

A Sustainable Innovation for Everyday Life

Unlike high-cost industrial systems, GreenCloset is designed to be affordable, accessible, and adaptable. Whether in a student hostel, a family home, or a luxury boutique, its modular build and intuitive alerts allow it to be deployed by anyone, anywhere. Importantly, the project encourages behavioral transformation — prompting users to become more aware of how they store clothes, how often they inspect them, and what they can do to preserve them better.

This small shift in user awareness, magnified across households and communities, can lead to a large-scale reduction in textile waste — one of the most overlooked environmental concerns today.

Redefining Smart Storage

GreenCloset isn't just a smart device — it's a lifestyle companion designed for a world that is waking up to the consequences of wastefulness. It embodies a vision where sustainability isn't an afterthought but a built-in feature of every product we design. Through intelligent sensing, real-time feedback, and user-friendly technology, GreenCloset redefines how we interact with our clothes and storage spaces. It's a step toward a future where wardrobes, too, are a part of the smart, sustainable home ecosystem — conscious, connected, and caring.

CHAPTER 2

PROBLEM STATEMENT

In many modern households, particularly those located in tropical and coastal regions, wardrobes and storage spaces are highly susceptible to fluctuating humidity levels and inadequate ventilation. These conditions create a microenvironment that is ideal for the proliferation of mold, mildew, and fabric deterioration. Despite the rapid adoption of smart home technologies across various domains—such as lighting, security, and energy management—there remains a notable gap in addressing this specific and recurring issue related to textile preservation and wardrobe hygiene.

Clothing and fabric-based items represent both monetary and sentimental value, yet their protection is often neglected. In the absence of a dedicated monitoring system, users frequently encounter premature damage, discoloration, unpleasant odors, and the irreversible degradation of garments. This not only results in an economic burden due to frequent replacement but also contributes significantly to the growing problem of textile waste, which exacerbates the environmental footprint of the fashion and garment industries.

The lack of an intelligent, responsive, and affordable solution that can proactively monitor, detect, and alert users about adverse environmental changes inside wardrobes is a pressing concern. Moreover, prolonged exposure to mold spores can pose serious health risks, especially for individuals with respiratory sensitivities, allergies, or weakened immune systems—making it a matter of public health as well.

There is a compelling need for a compact, user-friendly, and IoT-enabled system that continuously tracks temperature and humidity levels, provides real-time alerts, and promotes preventive action before damage occurs. Such a system can empower users to maintain the integrity, hygiene, and longevity of their clothing, thereby encouraging sustainable fashion practices and a more environmentally responsible lifestyle.

CHAPTER 3

OBJECTIVE

The core objective of GreenCloset is to conceptualize, design, and implement a smart, sustainable, and IoT-driven wardrobe monitoring system that can effectively detect and alert users to unfavorable environmental conditions—specifically humidity and temperature—that contribute to mold formation and textile degradation.

This intelligent system integrates a suite of sensors (such as DHT11/DHT22 for humidity and temperature sensing), a microcontroller (Arduino/ESP32), and real-time alert mechanisms (like buzzers, LEDs, or mobile notifications), forming a responsive and proactive framework for wardrobe management. By continuously monitoring ambient conditions within enclosed spaces like wardrobes, closets, and fabric storage units, GreenCloset empowers users to take timely preventive actions before mold or moisture damage occurs.

The case study's ambition extends beyond just preserving clothing. It emphasizes a larger vision of sustainability by reducing fabric waste, minimizing the frequency of clothing disposal, and decreasing reliance on fast fashion due to repeated purchases caused by mold-damaged garments. This directly contributes to reducing the environmental burden associated with textile production, dyeing, and disposal—activities known to be among the most resource-intensive and polluting in the world.

GreenCloset also strives to be:

- **Affordable:** Using cost-effective components to make the solution accessible for all socio-economic classes.
- **Modular and Scalable:** Adaptable across a wide variety of storage environments, from household wardrobes to boutique showrooms, travel luggage, and textile archives.
- **User-Centric:** With a simple interface and minimal learning curve, making it easy for users of all ages and tech backgrounds.

- **Energy Efficient:** Prioritizing low-power consumption to ensure sustainable operation even during extended monitoring.

Ultimately, GreenCloset aims to bridge the gap between smart living and sustainable practices—making intelligent wardrobe monitoring not just a luxury, but a practical and responsible lifestyle choice. It hopes to initiate a broader awareness of material care, responsible fashion consumption, and environment-conscious innovations in everyday life.

CHAPTER 4

SCOPE AND MOTIVATION

In today's increasingly technology-driven world, the integration of smart systems into everyday life has become more than a convenience — it's a necessity. One often overlooked yet impactful problem in household maintenance is excess humidity and mold formation inside wardrobes and enclosed storage spaces. Prolonged exposure to moisture can damage clothing, promote bacterial growth, and ultimately contribute to textile waste. This growing concern served as the primary motivation behind the development of GreenCloset — an IoT-powered smart wardrobe monitoring system.

GreenCloset aims to empower users with real-time environmental awareness by continuously monitoring key factors such as humidity and temperature, and instantly alerting them when values exceed mold-prone thresholds. This proactive approach helps preserve clothing quality, ensures better hygiene, and reduces the need for frequent replacements — aligning with both sustainable living practices and cost-effective wardrobe management. While its primary application targets household wardrobes, the system's modular design allows it to be adapted for a variety of other micro-environments — including lockers, closets, suitcases, museum textile archives, and commercial fashion storage. This flexibility makes GreenCloset not only a wardrobe care solution but a scalable framework for diverse environmental monitoring needs.

The scope of GreenCloset covers the design and implementation of a modular, affordable, and intelligent system using widely available components like the DHT11/DHT22 sensors for humidity and temperature, ESP32 or Arduino Uno as the microcontroller, and Wi-Fi or Bluetooth modules for seamless alert communication. The system is designed to be user-friendly, scalable, and adaptable to various use cases beyond household wardrobes — such as lockers, closets, suitcases, museums, and fashion retail storage.

Additionally, the system’s architecture leaves room for future expansion — including cloud data logging, AI-driven mold prediction, smartphone integration, and automatic ventilation triggers. These possibilities open a path for GreenCloset to evolve into a broader environmental monitoring platform, making enclosed spaces smarter, more efficient, and more sustainable.

In essence, GreenCloset merges technological innovation with daily-life practicality, addressing a simple problem with a powerful solution — and encouraging a shift toward smarter, healthier, and environmentally responsible storage habits.

Flowchart of GreenCloset

The flowchart of GreenCloset outlines the system’s logical sequence, showcasing how the smart wardrobe environment functions from startup to intelligent environmental monitoring and user alerting. This system is designed for real-time humidity and temperature tracking to protect garments from mold and moisture damage. Below is a thorough explanation of each step:

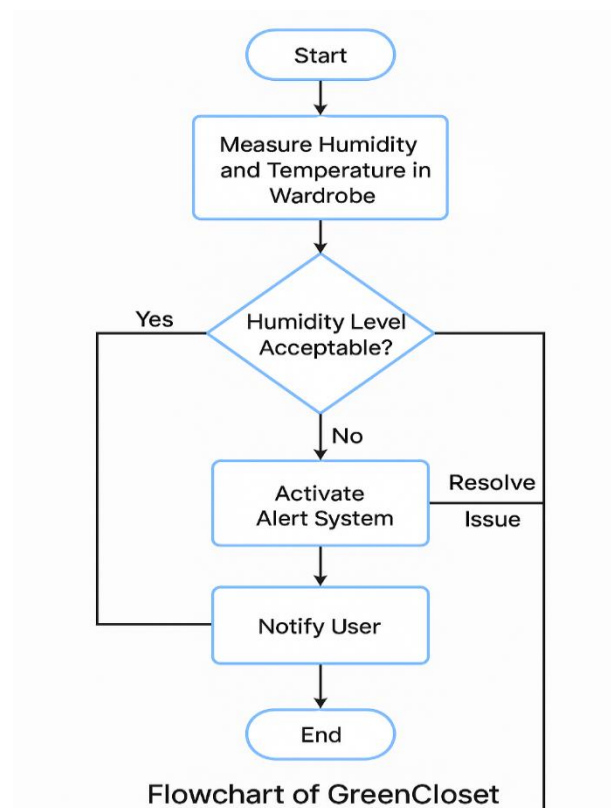


Fig.4.1 Flowchart of GreenCloset

1. System Initialization

At this first stage, the GreenCloset system is powered on, and the microcontroller (e.g., ESP32, Arduino Uno, or Raspberry Pi Pico) begins its initialization process.

- **Hardware Check:** The microcontroller checks the availability of essential components—like sensors, Wi-Fi modules, alert devices (buzzer or LED), and power supply.
- **Sensor Setup:** Sensors such as **DHT11** or **DHT22** are configured to start collecting humidity and temperature data.
- **Default Thresholds:** Predefined safe limits (e.g., 60% humidity) are loaded into the system memory.
- **System Readiness Message:** Optionally, an LED or serial monitor output indicates that the system is ready to operate.

This step ensures all components are functional and ready to begin data acquisition and monitoring.

2. Read Sensor Data

Once initialized, the system proceeds to gather real-time environmental data from sensors installed inside the wardrobe.

- **Humidity Sensor:** Continuously captures the percentage of moisture in the air.
- **Temperature Sensor:** Provides the ambient temperature which, when analyzed alongside humidity, gives better insight into mold risk.
- **Sampling Interval:** Data is typically read every few seconds or minutes, depending on power constraints and application.

This phase creates the raw data on which decisions are made.

3. Compare Data With Threshold Values

The core intelligence of the system lies in this step.

- The system evaluates whether the **humidity exceeds critical thresholds** (e.g., above 60% RH).
- If humidity is normal but the temperature is high, it may still present a risk when combined.
- This decision logic determines whether **current environmental conditions** pose a risk to stored textiles.

The comparison uses simple if-else logic, enabling rapid response with minimal computational load.

4A. If Safe – Continue Monitoring

If the data is within acceptable limits:

- The system goes back to reading new sensor data.
- No alerts are issued.
- LEDs may remain green (if visual feedback is provided).
- Power consumption remains low, ensuring energy efficiency.

This loop continues indefinitely until an unsafe condition is detected. This represents the steady state or "normal operation" of GreenCloset.

4B. If Unsafe – Trigger Alert Mechanism

If the sensor data shows values above the safe threshold: The alert mechanism is activated:

- LED turns red
- Buzzer sounds
- Mobile notification is sent (if connected via Wi-Fi/Bluetooth)
- Optional: SMS or email can also be configured

This alert immediately informs the user that the wardrobe environment is unsafe and needs attention.

5. Log the Data

Each time an alert is triggered, or even during normal cycles:

- The system logs temperature and humidity values.
- Logs can be stored in:
 - SD Card
 - EEPROM memory
 - Cloud platforms (e.g., Firebase, Google Sheets, or Thingspeak)

Logging supports:

- Trend analysis
- Predictive maintenance
- Eco-behavioral data tracking

This data also helps users understand when their wardrobes are most vulnerable—e.g., during monsoon or summer.

6. **User Intervention/Action**

Upon receiving the alert:

- The user is expected to respond proactively, such as:
 - Opening the wardrobe for airflow
 - Placing dehumidifying agents (e.g., silica gel or baking soda)
 - Switching on exhaust fans or ventilation modules
 - Moving the wardrobe to a drier area

In future versions of GreenCloset, this step may also include automated physical responses such as:

- Automatically opening air vents
- Switching on a small fan
- Triggering a connected smart home dehumidifier

7. **Resume Monitoring**

After the user takes corrective action:

- The system continues to monitor the conditions.
- Alerts will cease once conditions return to normal.
- New readings are constantly taken to ensure stability.

This loop ensures the system remains vigilant, and re-enters the cycle of continuous environmental protection.

This flowchart models GreenCloset as a proactive guardian of clothing and textiles stored in enclosed spaces. From real-time detection to intelligent alerting and user response, every component works in tandem to prevent mold, promote sustainability, and reduce textile waste.

Block Diagram of GreenCloset

The block diagram below illustrates the working of the Smart Humidity & Mold Detection System designed to monitor environmental conditions within wardrobes or enclosed storage spaces. The system employs sensor-based automation to detect high humidity levels and trigger an alert mechanism, thereby preventing mold growth and ensuring fabric longevity.

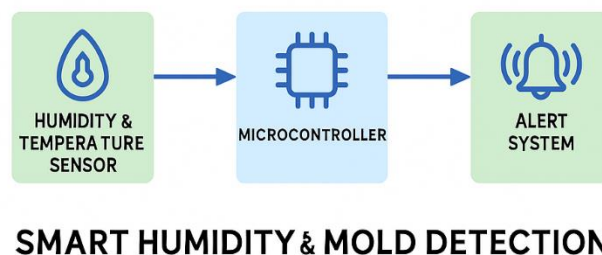


Fig.4.2 Block Diagram of GreenCloset

1. Humidity & Temperature Sensor

This is the first component of the system responsible for constantly monitoring the ambient humidity and temperature levels within the wardrobe or enclosure. Common sensors like the DHT11 or DHT22 can be used, which provide both humidity and temperature data.

- If the humidity crosses a predefined threshold (e.g., 60-65%), it sends a signal to the microcontroller.
- This helps in detecting early signs of mold-prone conditions.

2. Microcontroller

The microcontroller acts as the brain of the system, processing the data received from the sensor. Examples include Arduino Uno, ESP32, or Raspberry Pi Pico.

- It compares the sensor input with preset safe ranges.
- If the humidity is too high or sustained for long durations, it triggers the alert system.
- It may also store data or send it wirelessly to an IoT dashboard for monitoring.

3. Alert System

When abnormal humidity is detected, the microcontroller activates the alert system.

- This can include LED indicators, buzzers, or even smartphone notifications (via Wi-Fi modules or Bluetooth).
- Users are alerted in real-time, allowing them to take necessary action like opening the wardrobe, activating a dehumidifier, or removing damp clothes.

This smart system integrates sensing, processing, and alerting to provide a proactive solution for mold prevention in personal storage spaces. It ensures sustainable wardrobe maintenance, extending the life of clothes and maintaining hygiene.

CHAPTER 5

ADVANTAGES

GreenCloset offers a wide range of benefits by combining sustainability with smart technology. It not only protects clothing from environmental damage but also promotes eco-friendly wardrobe maintenance through automation and real-time monitoring. The system is designed to be affordable, energy-efficient, and adaptable to different storage settings, making it a practical solution for modern, conscious living.



Fig.5.1 Advantages of GreenCloset

Prevention of Mold and Fabric Damage:

GreenCloset constantly monitors microclimatic conditions within the wardrobe, particularly humidity and temperature, to detect mold-friendly environments. Early detection prevents microbial growth, safeguarding garments from irreversible damage and discoloration.

Promotes Sustainable Living:

By minimizing fabric deterioration and extending the usable life of clothes, the system contributes to reducing textile waste. It supports eco-conscious lifestyles by aligning with the principles of circular fashion and responsible garment care.

Real-Time Alerts and Actionable Feedback:

The system instantly notifies users when humidity levels exceed safe limits. These real-time alerts allow users to act promptly—be it ventilating the space, removing moisture sources, or activating dehumidifiers.

Affordable and Scalable:

Using widely available and cost-effective components like DHT11 sensors and microcontrollers, the system is budget-friendly. It can also be easily expanded or replicated for larger wardrobes, commercial storage, or industrial setups.

Energy-Efficient Operation:

Designed with low-power components, GreenCloset ensures continuous operation without burdening energy resources. Its efficient design supports long-term monitoring with minimal environmental impact.

Minimal Human Intervention:

The system functions autonomously once installed, eliminating the need for manual inspection. Users receive alerts without needing to constantly monitor conditions, thus saving time and effort.

Customizable and Upgradeable:

GreenCloset's modular architecture makes it easy to integrate new features, such as app-based controls, automated ventilation, or predictive analysis using AI, adapting to evolving user needs.

Applicable to Multiple Use-Cases:

Though initially designed for wardrobes, the system can be extended to protect other sensitive storage areas such as museum textile collections, shoe cabinets, libraries, and archival rooms.

Easy Installation and Maintenance:

The system is designed with simplicity in mind, allowing users to install it within their wardrobes without professional assistance. Its plug-and-play architecture and minimal wiring reduce complexity and ensure low maintenance over time.

Enhances Overall Hygiene and Air Quality:

By actively monitoring and controlling humidity, GreenCloset helps eliminate the musty odors and microbial buildup often found in closed storage spaces. This not only ensures that garments remain fresh and breathable but also improves the overall hygiene within the wardrobe environment.

CHAPTER 6

DISADVANTAGES

While GreenCloset presents a modern and eco-conscious solution for wardrobe care, certain limitations must be acknowledged to provide a balanced understanding of its practical implementation. These disadvantages help identify improvement areas and inform future enhancements to make the system more robust and user-friendly.



Fig.6.1 Disadvantages of GreenCloset

Initial Hardware Cost:

Though relatively affordable, the upfront cost of purchasing sensors, microcontrollers, and supporting components may be a barrier for users with budget constraints, especially when applied on a larger scale.

Limited to Humidity & Temperature Monitoring:

The current system does not detect other harmful conditions such as pest infestations, dust accumulation, or exposure to UV light—all of which can also damage clothing and stored fabrics.

Dependence on Wi-Fi or Power Supply:

For real-time updates and continuous performance, GreenCloset relies on uninterrupted power and a stable internet connection. In offline or low-connectivity areas, system efficiency may drop significantly.

No Automated Remedial Action:

While GreenCloset excels at detecting humidity fluctuations and notifying users, it lacks the capability to take immediate corrective actions on its own. Devices like dehumidifiers or exhaust fans must still be operated manually by the user, which may lead to delays in response if unattended. This limits the system's efficiency in preventing rapid mold growth in critical conditions.

Sensor Accuracy Constraints:

Inexpensive sensors like the DHT11 may show minor fluctuations or inaccuracies, particularly in rapidly changing humidity conditions, potentially affecting system reliability.

Risk of False or Frequent Alerts:

Small and temporary changes in the wardrobe environment may trigger frequent alerts, which over time can cause users to ignore warnings or disable notifications altogether.

Requires Basic Technical Knowledge:

Users without a technical background may face challenges in setting up, troubleshooting, or modifying the system, especially if any wiring or programming adjustments are needed.

Hardware Susceptible to Moisture Damage:

Ironically, if not well-insulated, some components may degrade due to prolonged exposure to the very humidity they are designed to monitor, leading to maintenance issues.

Data Privacy Concerns:

If cloud storage or mobile integration is added in future versions, users may be concerned about data usage, personal wardrobe analytics, and overall system security.

Aesthetic Disruption:

The integration of sensors, wiring, and electronic modules may interfere with the visual aesthetics of premium or minimalist wardrobe interiors. Without careful design or concealment, these components might appear intrusive or cluttered.

CHAPTER 7

APPLICATIONS

The versatility of GreenCloset lies in its wide range of real-world use cases that extend beyond simple wardrobes. With its smart sensing, real-time monitoring, and eco-conscious design, the system holds immense potential across multiple environments where moisture control is vital for the preservation of materials. Below is a detailed exploration of key domains where GreenCloset can make a meaningful impact:



Fig.7.1 Applications of GreenCloset

1. Household Wardrobes & Personal Closets

The most direct and significant application of GreenCloset is in everyday wardrobes found in residential settings. Many households—especially in humid regions—struggle with the gradual accumulation of moisture within enclosed wooden wardrobes or fabric-lined storage units. Over time, this can lead to mold outbreaks, musky odors, and irreversible garment damage. GreenCloset serves as a vigilant sentinel, continuously tracking environmental parameters and issuing alerts before damage can occur. This proactive defense not only extends

the life of clothes but also minimizes the need for repeated dry cleaning or replacements, aligning with sustainable living practices.

2. Coastal & Tropical Region Storage Solutions

Geographies with high atmospheric moisture—such as tropical or coastal towns—present a chronic challenge when it comes to clothing storage. Even high-end wardrobes and closets in these regions are susceptible to mold, mildew, and corrosion of metallic accessories like zippers and hooks. GreenCloset can be installed in such regions as a cost-effective humidity watchdog, especially useful in households that do not use air conditioning round the clock. Its ability to work passively in the background makes it a valuable companion for long-term clothing care, particularly during monsoon seasons or in beach-facing homes.

3. Fashion Boutiques & Retail Showrooms

Retail environments that house premium clothing collections face a significant risk of quality degradation due to storage in closed racks or display cabinets. Moisture-related issues can silently affect the texture, color, and even scent of clothes on sale. Integrating GreenCloset into such spaces allows store owners to maintain a controlled micro-environment, assuring customers of fresh, damage-free garments. It can also help avoid losses associated with unsellable stock and reduce customer complaints, thereby improving brand credibility and sales reliability.

4. Museum Textile Archives & Heritage Storage

Museums that preserve historic clothing, tapestries, and textiles have extremely strict climate control requirements. Even the slightest exposure to high humidity can lead to fabric deterioration, mold, or color bleeding—damaging irreplaceable heritage artifacts. GreenCloset offers a scalable and unobtrusive solution for monitoring these delicate environments. It can be implemented inside display cases or archival drawers, ensuring preservation without interfering with visual aesthetics. The system's data logs can further help

archivists monitor long-term storage trends and take preventive conservation measures.

5. Travel Luggage & Portable Storage

Frequent travelers often encounter challenges when clothes remain sealed inside suitcases for extended durations—especially in damp weather conditions or during international journeys. Moisture trapped inside luggage can lead to unpleasant odors or wrinkling of garments. A compact version of GreenCloset can be embedded into modern suitcases or travel trunks, offering smart humidity tracking and app-based alerts while on the move. Such an integration can elevate the value of premium luggage brands while solving a highly relatable travel problem.

6. Small Business Warehouses & Tailor Shops

Local tailoring units, boutique studios, and small-scale warehouses often lack sophisticated HVAC systems and climate control infrastructure. However, these spaces still house large volumes of clothing and fabric that are highly susceptible to humidity damage. Installing GreenCloset can empower these businesses with a budget-friendly monitoring solution, reducing textile spoilage and enhancing customer satisfaction. Over time, such a system can also support efficient inventory management by categorizing garments based on optimal storage requirements.

7. Laundry Rooms & Drying Areas

Laundry rooms often accumulate high levels of ambient humidity due to drying clothes, steam ironing, or washing machine exhausts. This makes them a hotspot for mold and mildew growth, not only on fabrics but also on walls and wooden cabinets. GreenCloset can be placed in such areas to offer real-time tracking and alert homeowners before moisture levels cross dangerous thresholds. When integrated with automated ventilation or dehumidifiers, the system can help create a cleaner, safer, and mold-free laundry space.

8. School, Hostel & Gym Lockers

Lockers in educational institutions and fitness centers often store damp clothes, sports uniforms, or shoes—creating the perfect environment for microbial growth. Since these lockers are typically closed and rarely cleaned, mold can quickly spread and even impact indoor air quality. GreenCloset can be modified into a miniaturized version suitable for individual lockers, offering students or gym-goers a hygienic storage solution. It can also be integrated into central locker management systems for facility maintenance staff.

9. Theatrical Costume Rooms

Theatre production houses and film studios frequently manage large wardrobes of costumes, many of which are delicate, expensive, or historically inspired. These are stored for long durations and reused across performances or shoots. Humidity-induced damage to such garments can lead to production delays and cost overruns. Implementing GreenCloset can help costume designers and production managers preserve outfit quality while also ensuring actor safety through clean and odor-free costumes.

10. Textile Research Labs & Fabric Testing Units

In academic or commercial textile research environments, controlling microclimates is crucial for testing fabric durability under different conditions. GreenCloset can be deployed as a testbed tool to simulate humidity fluctuations and observe their effects on various textiles. This adds value not only in material research but also in product development for industries like activewear, military uniforms, or eco-friendly fabrics.

CHAPTER 8

CONCLUSION

The convergence of sustainability and technology is no longer a distant ideal—it's a tangible, evolving reality. GreenCloset exemplifies this shift by addressing a subtle yet significant environmental and lifestyle concern: the preservation of wardrobes against humidity-induced mold. By leveraging the potential of IoT, the system transforms traditional storage spaces into intelligent environments that are capable of sensing and responding to their conditions. It fosters a culture of conscious living where clothing longevity, resource conservation, and user awareness are harmoniously interwoven.

This case study goes beyond simple environmental monitoring; it represents a step toward preventive care in daily life. Often, damage caused by humidity and mold goes unnoticed until it's too late, resulting in irreversible textile deterioration and increased waste. GreenCloset intervenes before such outcomes occur—alerting users promptly and encouraging timely actions, thereby aligning itself with the principles of environmental mindfulness and responsible consumption.

While the current iteration of GreenCloset primarily focuses on monitoring and alerting, the scope for future integration with automated control systems opens doors to a more self-sufficient and proactive wardrobe ecosystem. With enhancements such as smart ventilation, cloud-based analytics, and user-centric design, this system holds potential to scale both in functionality and impact.

Ultimately, GreenCloset is not just a technological initiative—it's a lifestyle innovation. It embodies the future of smart living by protecting what we wear, conserving what we store, and contributing in its own subtle way to a more sustainable and intelligent world.

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